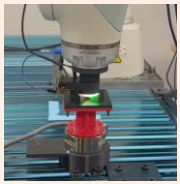
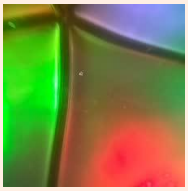
(a) Register real contacts

One registered pose per label



Registered press

q^*



Real image

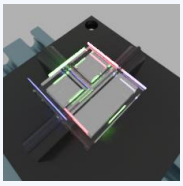
+



Registered depth

(b) Simulate contacts

Renderer fitting + appearance gap

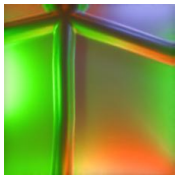


Simulated contact



Depth target

render



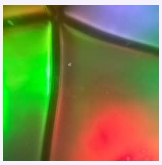
PBR



Learned

(c) AlignTac: real appearance + aligned geometry

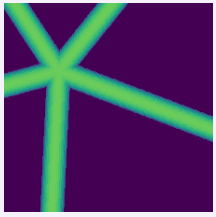
Unlabeled real image





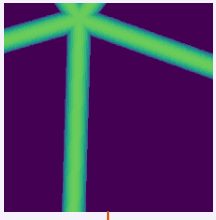
Known simulated geometry

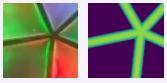
Initial geometry



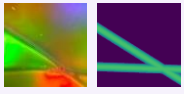
AlignTac

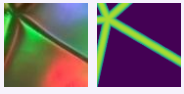
Aligned geometry





Few registered touches







Many pseudo-labeled contacts for supervision